Terraform Task

Task Description:

Launch Linux EC2 instances in two regions using a single Terraform file.

Step 1 : Configured aws cli in terraform server.

```
ubuntu@ip-172-31-40-191:~$ aws configure
AWS Access Key ID [None]: AKIA4F35FI6UL3HODB6O
AWS Secret Access Key [None]: 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
Default region name [None]: us-west-2
Default output format [None]:
```

Step 2 - Created a terraform script to launch instances in N. Verginia and Mumbai

```
terraform {
 required_providers {
  aws = {
   source  = "hashicorp/aws"
   version = "~> 5.0"
  }
 }
}
provider "aws" {
 alias  = "us_east_1"
 region = "us-east-1"
```

```
}

provider "aws" {
  alias  = "ap_south_1"
  region = "ap-south-1"
}

data "aws_ami" "amazon_linux_us" {
  provider    = aws.us_east_1
  most_recent = true
  owners      = ["amazon"]
  filter {
    name   = "name"
    values = ["al2023-ami-*-x86_64"]
  }
}

data "aws_ami" "amazon_linux_in" {
  provider    = aws.ap_south_1
  most_recent = true
  owners     = ["amazon"]
  filter {
    name   = "name"
    values = ["al2023-ami-*-x86_64"]
  }

resource "aws_instance" "us_ec2" {
  provider      = aws.us_east_1
  ami          = data.aws_ami.amazon_linux_us.id
```

```
  instance_type = "t2.micro"


  tags = {
    Name = "linux-us-east-1"
  }
}
resource "aws_instance" "in_ec2" {
  provider      = aws.ap_south_1
  ami           = data.aws_ami.amazon_linux_in.id
  instance_type = "t2.micro"
  tags = {
    Name = "linux-ap-south-1"
  }
}
```

Step 3 - Executed the script

terraform init

```
ubuntu@ip-172-31-40-191:~/project$ terraform init
Initializing the backend...
Initializing provider plugins...
- Finding hashicorp/aws versions matching "~> 5.0"...
- Installing hashicorp/aws v5.100.0...
- Installed hashicorp/aws v5.100.0 (signed by HashiCorp)
Terraform has created a lock file .terraform.lock.hcl to record the provider
selections it made above. Include this file in your version control repository
so that Terraform can guarantee to make the same selections by default when
you run "terraform init" in the future.
```

terraform plan

```
ubuntu@ip-172-31-40-191:~/project$ terraform plan
data.aws_ami.amazon_linux_us: Reading...
data.aws_ami.amazon_linux_in: Reading...
data.aws_ami.amazon_linux_us: Read complete after 1s [id=ami-02f058bc6b3ee6fcc]
data.aws_ami.amazon_linux_in: Read complete after 2s [id=ami-02a8bcaaf6d87b759]

Terraform used the selected providers to generate the following execution plan. Resource actions are indicated with the following symbols:
  + create

Terraform will perform the following actions:

  # aws_instance.in_ec2 will be created
  + resource "aws_instance" "in_ec2" {
      + ami                                  = "ami-02a8bcaaf6d87b759"
      + arn                                  = (known after apply)
```

terraform apply

```
ubuntu@ip-172-31-40-191:~/project$ terraform apply
data.aws_ami.amazon_linux_us: Reading...
data.aws_ami.amazon_linux_in: Reading...
data.aws_ami.amazon_linux_us: Read complete after 1s [id=ami-02f058bc6b3ee6fcc]
data.aws_ami.amazon_linux_in: Read complete after 2s [id=ami-02a8bcaaf6d87b759]

Terraform used the selected providers to generate the following execution plan. Resource actions are indicated with the following symbols:
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Terraform will perform the following actions:

  # aws_instance.in_ec2 will be created
  + resource "aws_instance" "in_ec2" {
      + ami                                  = "ami-02a8bcaaf6d87b759"
      + arn                                  = (known after apply)
```

```
Do you want to perform these actions?
  Terraform will perform the actions described above.
  Only 'yes' will be accepted to approve.

  Enter a value: yes

aws_instance.us_ec2: Creating...
aws_instance.in_ec2: Creating...
aws_instance.us_ec2: Still creating... [00m10s elapsed]
aws_instance.in_ec2: Still creating... [00m10s elapsed]
aws_instance.us_ec2: Creation complete after 14s [id=i-0229e4cec05fb7ac5]
aws_instance.in_ec2: Creation complete after 15s [id=i-04feccd7de1c5ee2e]

Apply complete! Resources: 2 added, 0 changed, 0 destroyed.
```

Instances started running in two regions